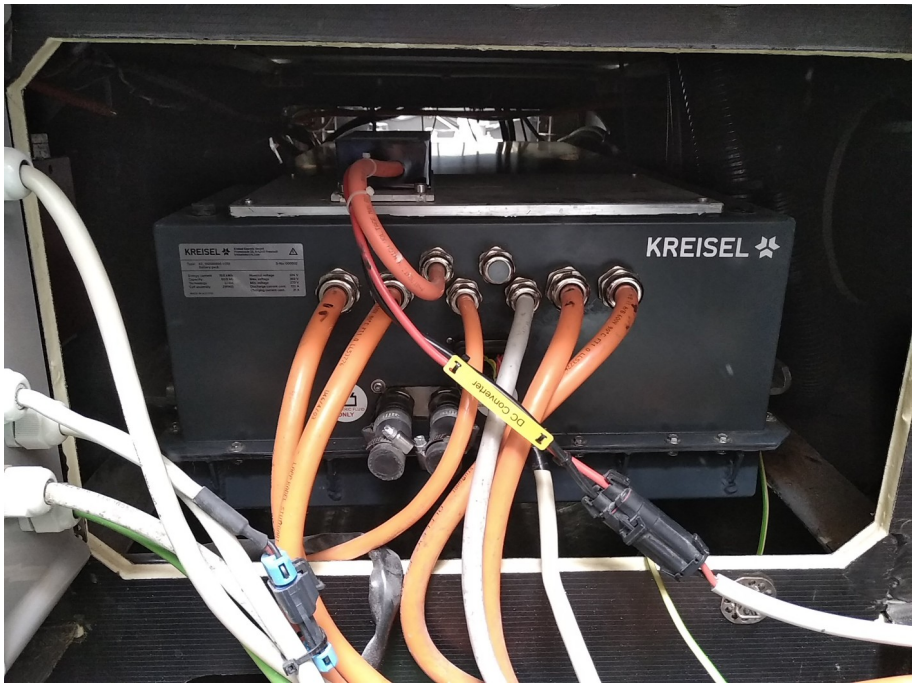
Kreisel High Voltage Battery

Diagnostic Log

Table of content

1 Version history....2
2 Equipment required....2
3 Introduction – Planning the process, spread over two days....3
3.1 No high voltage handling required....3
3.2 A four step process spread over two days....3
4 Setting up the software prior to getting to the boat....3
5 DAY 1 Session 1 : Charging the battery fully up....5
5.1 Switching the boat on....5
5.2 Retrieving boat software version....5
5.3 Charging the battery....6
6 DAY 1 Session 2 : Disconnecting the battery control connector at end of charge....7
- Switch the Iguana system off....7
- Accessing the battery....7
- Disconnecting the battery control connector....9
7 DAY 1 Session 3 : Preliminary check and 1st recording....10
7.1 Setting-up the hardware on the boat....10
- Plugging-in the diagnostic harness, CANbus module and laptop....10
7.2 Preliminary check (optional as might already have been done)....11
- What to check ?....11
- IMS software version required for the preliminary check....11
- Where to read the parameter values ?....12
7.3 Recording the data broadcast on the communication port at end of charge + 5h....12
7.4 Re-disconnecting the battery control connector....14
8 DAY 2 : 2nd recording (at end of charge + 29 hours, ie 24 hours after 1st recording)....16
8.1 Reconnecting the battery control connector....16
8.2 Recording the data broadcast on the communication port at end of charge + 29h....16
8.3 Removing diagnostic harness and returning to initial setup....16
8.4 Emailing files to Iguana Yachts....16
9 Post-processing files (for Iguana internal use)....17



1 Version history

- V1 : Initial version 6/1/2021
- V2 : Merged preliminary check and CANtrace recording. 17/2/2021
- V3 : Added file content checks. 23/3/2021

2 Equipment required

- Laptop computer with Ixxat drivers and canAnalyser3Mini software tool installed.
- Ixxat USB to CANbus interface module.
- Battery control connector interface harness. (for diagnostic purposes only).
- Shore power accessible so as to put boat on charge.

3 Introduction – Planning the process, spread over two days

3.1 No high voltage handling required

- Despite the battery being a high voltage battery (324V nominal voltage, 378V maximum cut-off voltage), the present diagnostic does not require to work on the battery high voltage part.

  The diagnostic is done through one of the two communications port of the battery, and as such, only requires to unplug and replug the low voltage (12V supplied) control connector of the battery.

3.2 A four step process spread over two days

- The principle of the diagnostic is to record two short lapses of battery data broadcast at 24 hours interval, the battery being asleep inbetween.
- Four steps :

  1) **DAY1 Session 1** : Put the boat on charge. Leave to charge fully up.
  2) **DAY 1 Session 2** : At end of charge, disconnect the battery control connector. Leave battery fully asleep for 5 hours.
  3) **DAY 1 Session 3** : Reconnect battery control connector, wake battery up for a one minute broadcast recording on one of its communications ports. Disconnect control connector again and leave to battery asleep for another 24h.
  4) **DAY 2** : Reconnect battery control connector, wake battery up for a one minute broadcast recording on one of its communications ports.

- Eventually, those recordings should be emailed to Iguana Yachts for diagnostic.



Process at boat should be initiated at a date and time that will match :

- Going to the boat to launch battery charge.
- Being at the boat at end of charge*.
- Going to the boat at end of charge + 5h.
- Going to the boat at end of charge +5h + 24h.

* Remaining charge time can be estimated as mentioned in § Day 1 Session.

4 Setting up the software prior to getting to the boat

- So as to interface a laptop computer to the battery CANbus network to be monitored.
- A CANbus to USB interface module similar to the one shown on the photo hereunder is necessary.

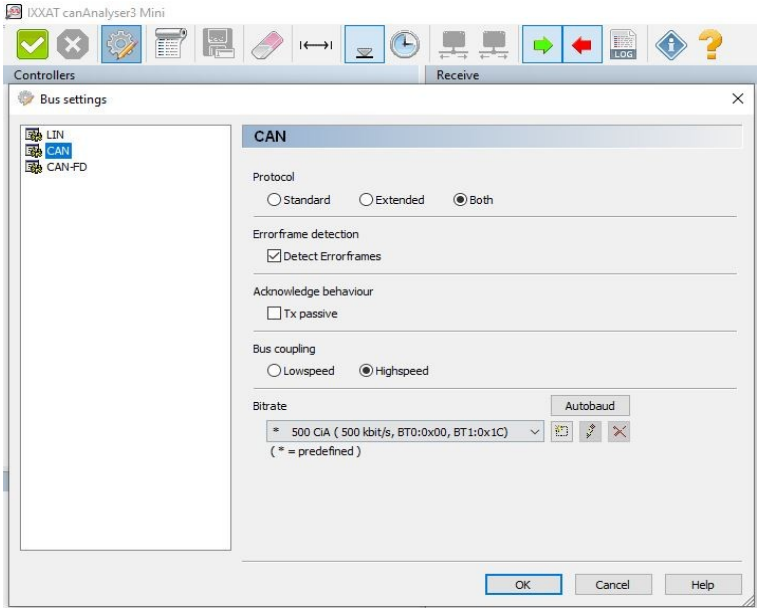
- On Commuters, it is supplied within the ex-factory boat kit.
- On Originals, it is supplied as an optional servicing accessory.



- Have the IXXAT interface module drivers installed onto a laptop computer : Ixxat VCI Setup 4.0.925.0.exe installation, followed by installation of the add-on VCI V2 API-Addon (VCI2 on VCI4)_2_31_133_0.exe.

  That will also install the canAnalyser3Mini application on the laptop.

- Launch the canAnalyser3Mini application on the laptop and set its settings (gearwheel icon) as shown hereunder. That application will allow to read and record CANbus communications from the battery.



5 DAY 1 Session 1 : Charging the battery fully up

5.1 Switching the boat on

- Switch the boat on :
  - On Commuters (cabin Iguana)
    - Switch the boat on from the I button to starboard of the steering wheel.

      The Iguana display should turn on. You do not have to start the track system (“Tracks” button).
  - On Originals (center console Iguana)
    - Switch the boat on from the switch key located above the outboard starting key.

      The Iguana display should turn on. You do not have to start the track system (F1 “Start” button).

5.2 Retrieving boat software version

- In anticipation of the optional preliminary check that will be done later once the battery is fully charged, the Iguana Mobility System (IMS) software version should be noted.

  Running the optional preliminary check requires the IMS software version to be :

  - On Commuters :
    - V23 softwares : V23.33 or newer.
    - V25 softwares : V25.31 or newer.

    This can be checked in Main menu  (Burger like button at the top left hand corner of the display) / System / Info on the Iguana IMS display.
  - On Originals :
    - V16.43 or newer.
    - Any V20.x

    This can be checked in Main menu  (Burger like button at the bottom right hand corner of the display) / Info on the Iguana IMS display.
- If your Iguana software version is older, please inform Iguana Yachts so that a software update can be carried out. This can be done at a later stage, even once the present diagnostic is fully complete. All that would then be needed would be to return to the boat for software update download.

5.3 Charging the battery

- Plug the shore power cord to a wall socket so as to put the boat in charge.
- Allowing about 15 seconds after plugging in shore power, check that the high voltage traction battery is charging : the battery vertical bargraph on the Iguana display should turn green and show a filling up motion.

  If not the case :

  - On Commuters
    - Check what the battery charge level is. This is called SOC (state of charge) : SOC is shown in percent (%) on the Battery HT field on the info page (accessed by pressing the  button at the top of the display).
    - If battery SOC is below 95%, then check the shore power, as the charger should be charging up the battery.
    - If battery SOC is 95% and above, the charger may not kick in (because of preset charge hysteresis). On Commuters with software V25.21 or newer, you can force it to charge by going to Main menu  / Settings / Master features / 4158 / Force Charging, and set this to True.

      If Force Charging is not available, then, so as to get the charger to charge again to 100%, the boat should be driven so as to draw battery energy down to less than 95% SOC. And then the shore power should be reconnected to launch a charge.
  - On Originals
    - Check what the battery charge level is (SOC : state of charge) : SOC is shown in percent (%) at the top of the display when in the startup menu.
    - If battery SOC is beyond 95%, the charger may not kick in (because of preset charge hysteresis). There is no way to override this from the display. To get the charger to charge again to 100%, the boat should be driven so as to draw battery energy down to less than 95% SOC. And then the shore power should be reconnected to launch a charge.
    - If battery SOC is not at 100%, then check the shore power, as the charger should be charging up the battery.
- Remaining charge time (in decimal hours) can be estimated from :

  Remaining charge hours = 187 x (100- SOC) / (Shore power voltage x Charger input Current)

  where :

  - SOC = Battery state of charge, in %.
  - Shore power voltage, in V.
  - Charger input current, in A, as set on the system :

- on Commuters :
    - V23 softwares : in Main menu / Settings / Master Battery / 4158 / Max Current IP
    - V25 softwares : in Main menu / Settings / Setup Charger / Charging Current
- On Originals : in Main menu / Settings / Setup / Max current line.

- Once the battery is fully charged, proceed to the next step.

6 DAY 1 Session 2 : Disconnecting the battery control connector at end of charge

Switch the Iguana system off

- Switch the boat off :
    - **On Commuters (cabin Iguana)**
        - On Commuters, the boat switches itself off once charging is complete. If not effectively switched off (Iguana display off), do so manually as follows.
        - Switch the boat off preferably from the OFF button at the top right hand corner of the Iguana display.

          The Iguana display should turn off.
        - Alternately, the system can be switched off from the 0 button to starboard of the steering wheel.
    - **On Originals (center console Iguana)**
        - Switch the boat off from the switch key located above the outboard starting key.

          The Iguana display should turn off.

Accessing the battery

- **On Commuters (cabin Iguana)**
    - Check that the boat is switched off (the Iguana display on the dashboard should be off).
    - In the cabin, the battery lies behind the cabin steps.

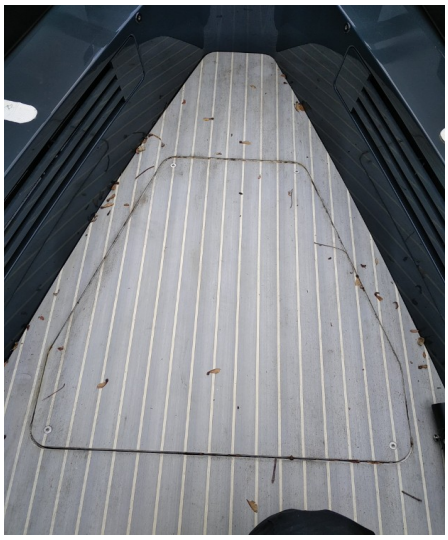
      The steps are clipped onto their supporting frame. Lift off the steps by grabbing the steps nosing and lifting up. The holding clips should release and let the steps lift off as a whole assembly.




- You will then be facing the battery front panel.



- **On Originals (center console Iguana)**
  - Check that the boat is switched off (the Iguana display on the dashboard should be off).
  - The battery is located below the steering centre console. Since we only have to access the battery front panel, access will be gained through the fore deck floor hatch.
  - Remove that fore deck floor hatch, retained by four allen head screws.

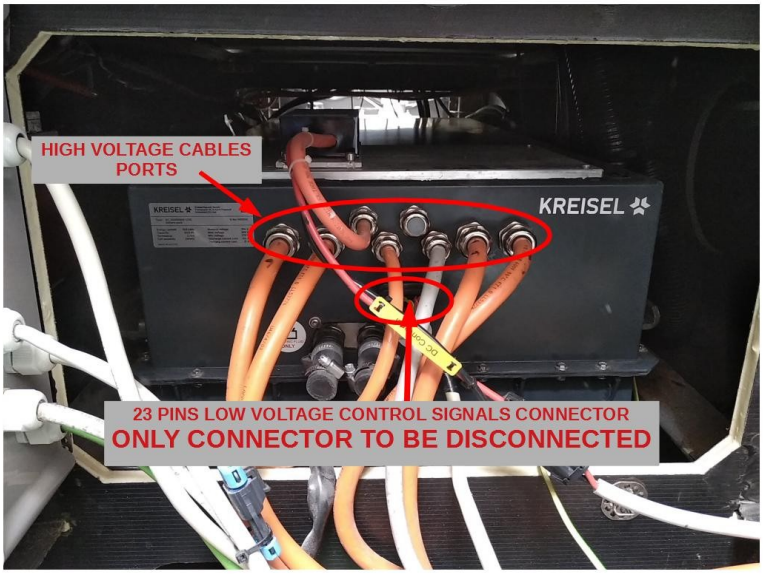


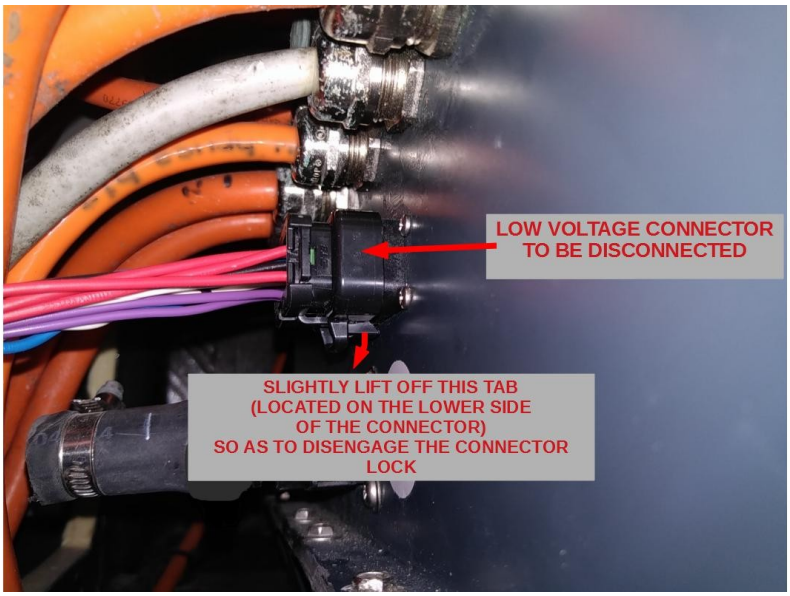

  - The battery front panel is accessible from the rear of that front bilge compartment.

Disconnecting the battery control connector

- The battery control port we have to access to, is on the battery front panel, as shown on the photos below.

Safety notice : Before proceeding, check again that the boat is switched off (Iguana display off).







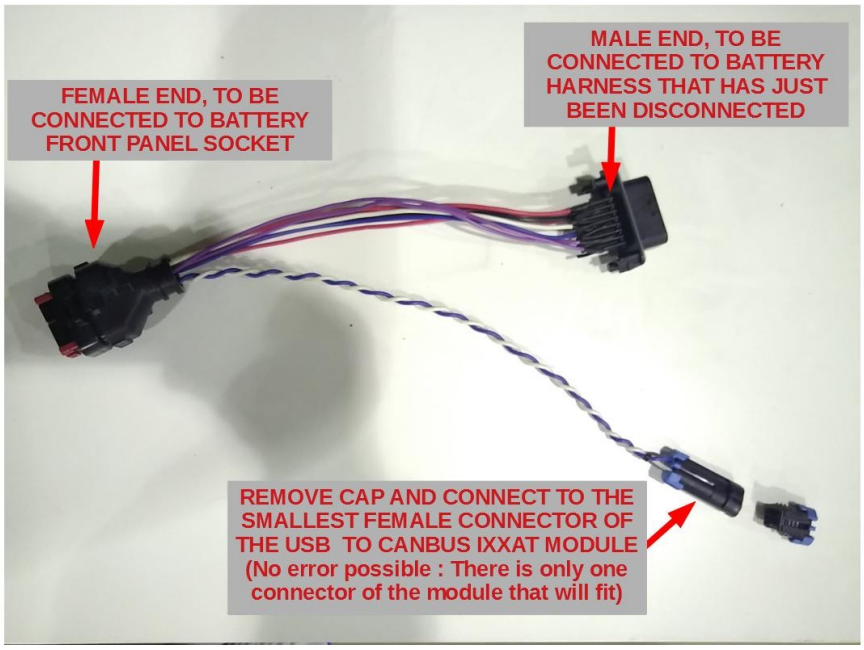
- Leave the battery disconnected until 5 hours have elapsed since end of charge, and **plan to come back by then.**

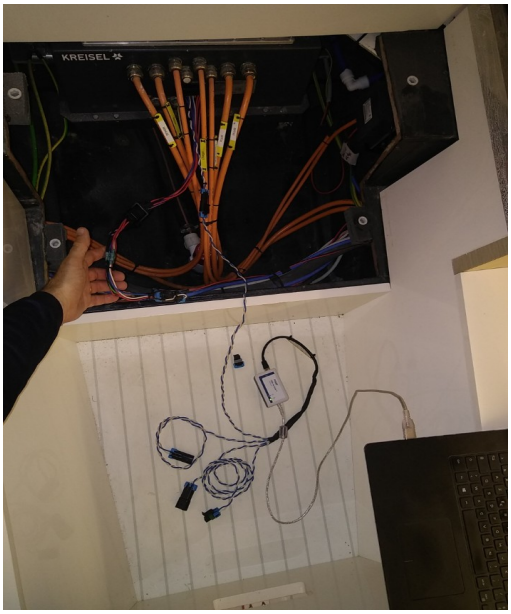
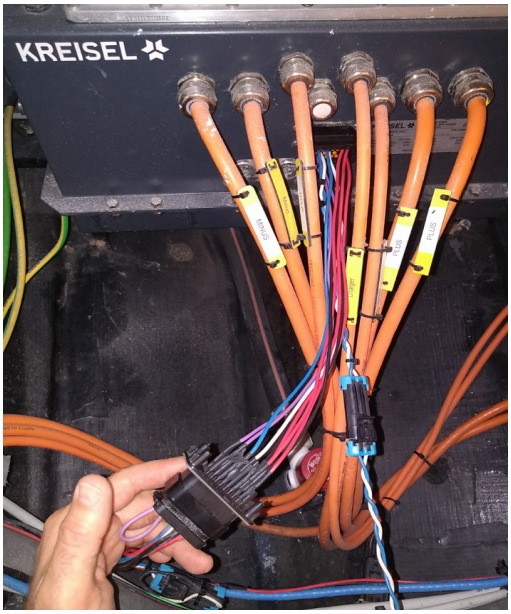
7 DAY 1 Session 3 : Preliminary check and 1st recording

7.1 Setting-up the hardware on the boat

Plugging-in the diagnostic harness, CANbus module and laptop

- The diagnostic harness should be plugged in series (in a daisy chain) inbetween the female connector that has been disconnected from the battery at end of charge, and the male connector socket on the battery front panel.



- The 2 way male connector at the end of the blue and white twisted wire pair should be connected to the mating female connector on the USB to CANbus IXXAT module. There are four connectors at the ends of the IXXAT module harness, but only one will match the diagnostic harness one. The three others should be left not connected.
- Plug the IXXAT module USB plug into any of the laptop USB ports.

7.2 Preliminary check (optional as might already have been done)

What to check ?

- A preliminary check can be performed by reading some parameter values on the Iguana display. It is not absolutely necessary as the battery broadcast recordings will carry that information, but this reading gives a first immediate indication on the battery state of health.

  The battery is composed of over a thousand individual cells. The minimum and maximum voltage amongst all those cells is indicated on one of the Iguana display submenus. It is those two values that we want to take note of.

IMS software version required for the preliminary check

- Switch the Iguana system on. The Iguana display should turn on.
- For those two voltage values to be shown on the Iguana display, the IMS software version should be :
  - On Commuters :
    - V23 softwares : V23.33 or newer.
    - V25 softwares : V25.31 or newer.

    This can be checked in Main menu (Burger like button at the top left hand corner of the display) / System / Info on the Iguana IMS display.
  - On Originals :

- V16.43 or newer.
- Any V20.x

This can be checked in Main menu (Burger like button at the bottom right hand corner of the display) / Info on the Iguana IMS display.

- If the software version presently installed on your Iguana is older than that, then we just have to download an update first. That can be done wirelessly using the IQANgo app from a smartphone, either in standalone or through simultaneous remote connection to our laptop. It is all just a matter of a few minutes. This can be done on any of the Day 1 or Day 2 four sessions.

Where to read the parameter values ?

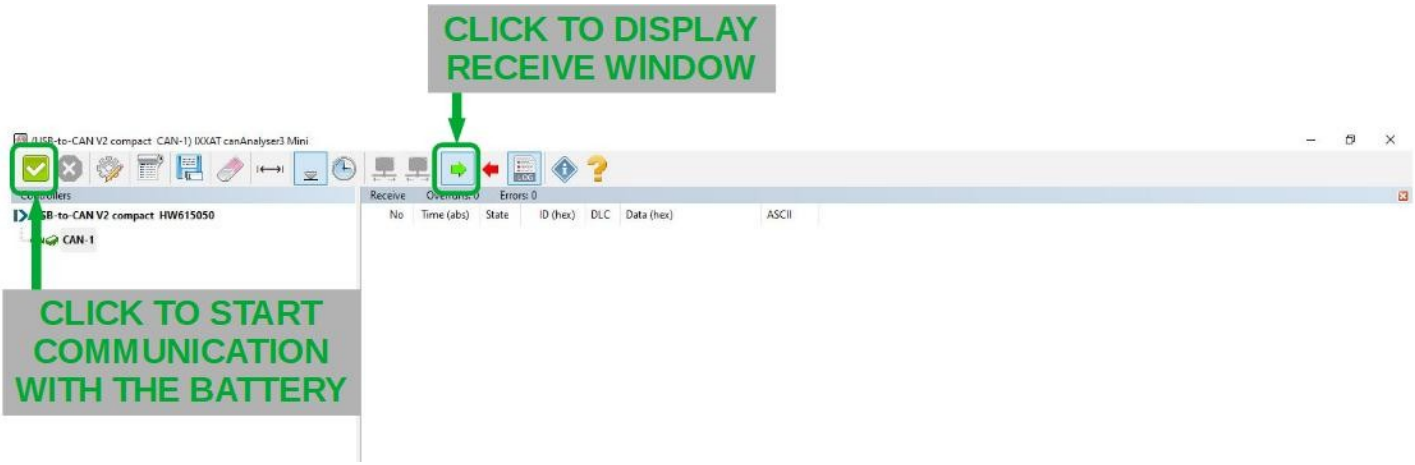
- On Commuters (cabin Iguana)
  - In Main menu / Data / Battery, scroll down the list until you get to the LowCellVoltage and HighCellVoltage lines.
  - Take a note or screen shoot those two values, and **please kindly forward them to Iguana Yachts for reference.**
- On Originals (center console Iguana)
  - In Main menu / Info, scroll down the list with the down arrow until you get to the LowCellVoltage and HighCellVoltage lines.
  - Take a note or screen shoot those two values, and **please kindly forward them to Iguana Yachts for reference.**

7.3 Recording the data broadcast on the communication port at end of charge + 5h

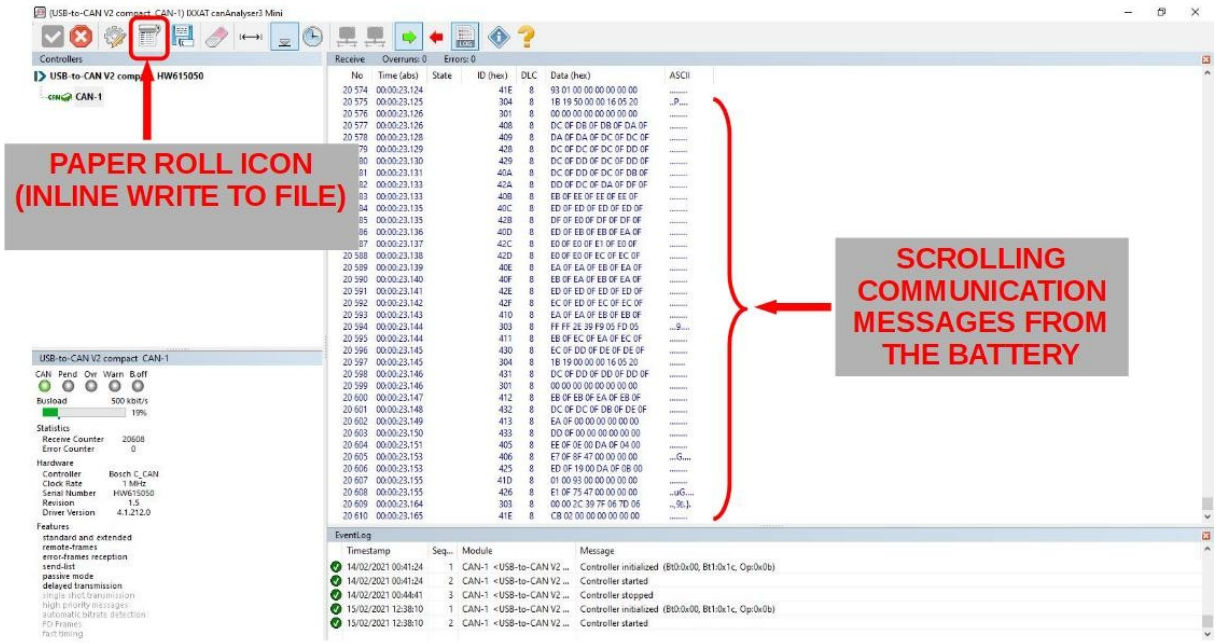
- If not already the case, switch the boat on :
  - **On Commuters**
    - Switch the boat on from the I button next to the steering wheel.

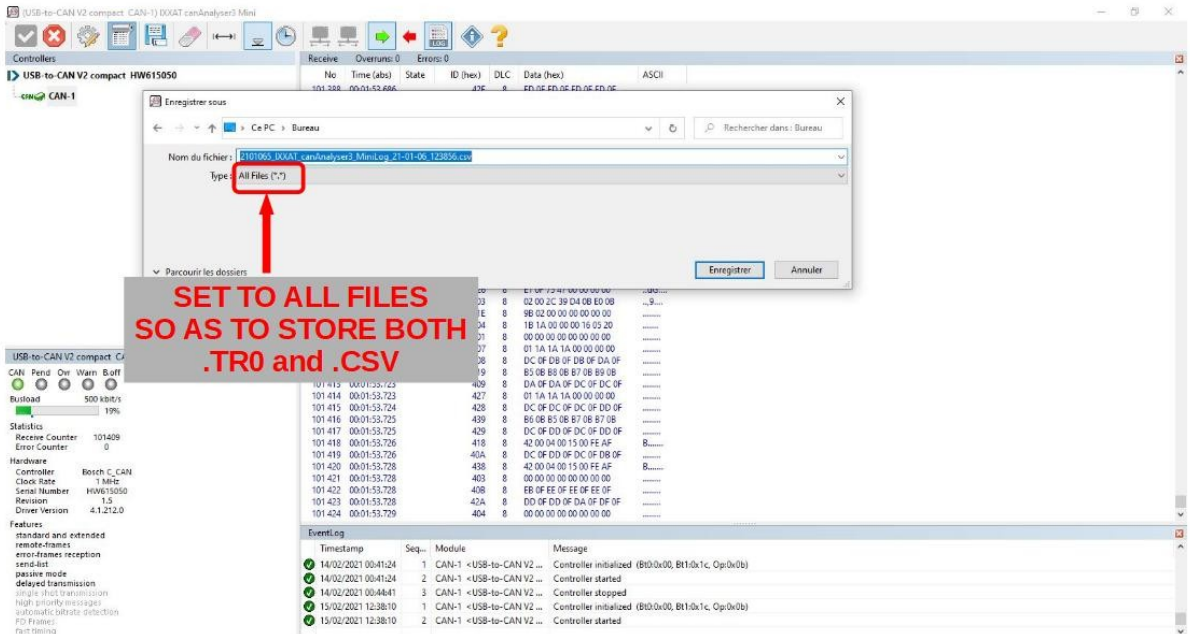
      The Iguana display should turn on. You do not have to start the track system (“Tracks” button).
  - **On Originals**
    - Switch the boat on from the switch key located above the outboard starting key.

      The Iguana display should turn on. You do not have to start the track system (F1 “Start” button).
- Launch the canAnalyser3Mini application on the laptop. Its settings have been previously set (in § “Setting up the software prior to getting to the boat”).
- Check that it is the Receive window which is active (Green arrow icon boxed/selected).
- Click on the "start all communications" icon (white tick in green box).



- You should then see lines of digital messages scroll down the receive window of canAnalyser3 on the laptop screen.
- If so, the setup is ready for recording the battery communication messages.
- Click on the "inline write to file" icon (paper roll icon). Choose a directory where to store the file, set file name (without extension) and set « file type » to all, so that both .csv and .tr0 are recorded. Click on Save.





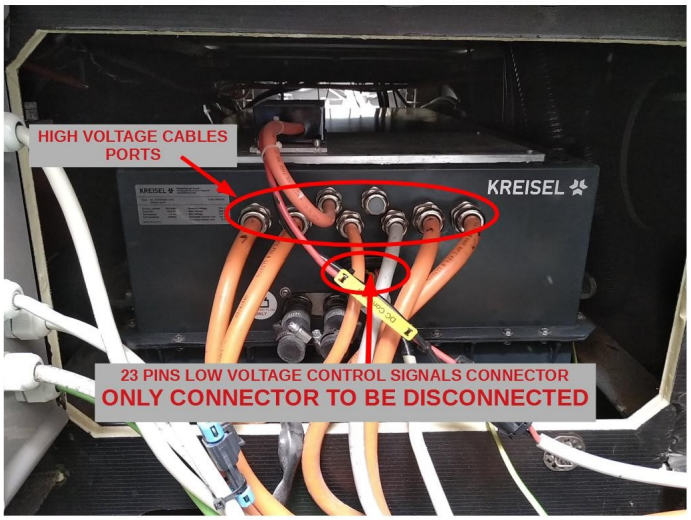
- Leave the application to record **for about one minute.**
- Then stop the recording by clicking on the “stop all communication” icon (white cross in red box).
- The recording is now over, with recording files stored.
- Check that both .tr0 and .csv file formats have been stored on your laptop (the “tr0” file is mandatory) and **check they are filled-up with data.** They should be over 1Mbyte in size for one minute of recording (about 1.5Mbytes/mn for the .tr0, 6Mbytes/mn for the .csv).
- Open the .csv file with a text editor, and check :
  - at the top of the file that bitrate is mentioned to be 500 kbits/s.
  - In the file text that there is no occurrence of the word “error”. If so, retry the recording or contact Iguana with the file as reference.

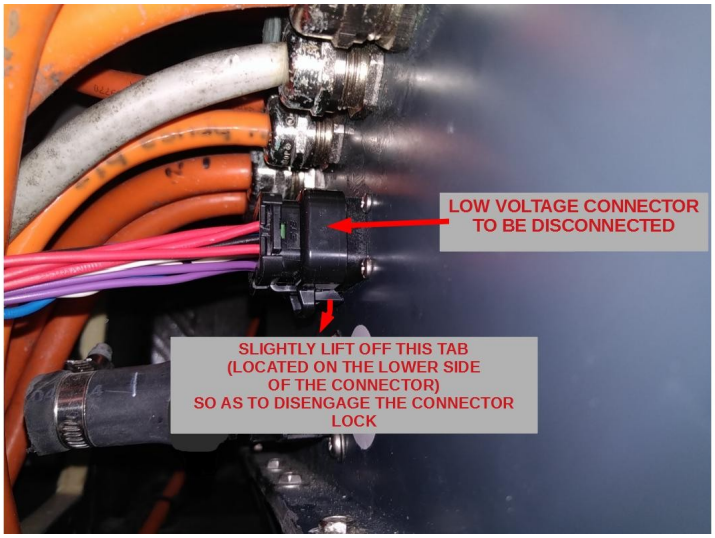
7.4 Re-disconnecting the battery control connector



Safety notice : Before proceeding, check again that the boat is switched off (Iguana display off).

- At this stage, disconnection can be done either at the battery front panel connector, or at the battery harness connector (where it connects to the diagnostic harness). In this latter case, it should be ensured that the IXXAT module also gets disconnected from the laptop or the battery (to avoid connection from the battery to the laptop USB port).







- Leave the battery disconnected for another 24 hours, ie until end of charge + 29h, and **plan to come back by then.**

8 DAY 2 : 2nd recording (at end of charge + 29 hours, ie 24 hours after 1st recording)

8.1 Reconnecting the battery control connector

- Re-plug the connections that were undone after first recording, namely : battery control connector, and potentially IXXAT module connection between battery and laptop.

8.2 Recording the data broadcast on the communication port at end of charge + 29h

- Proceed to a new 1 minute recording exactly in the same way as done on DAY 1 Session 3 : switch boat system on, launch CanAnalyser3Mini, set storage file, record for one minute, stop recording, check proper storage of both file formats, switch boat off.

8.3 Removing diagnostic harness and returning to initial setup

- Ensure the boat is switched off.
- Disconnect the diagnostic harness from battery front panel and battery harness.
- Re-plug battery harness into battery front panel connector. Ensure the connector is properly clicked in (click heard and/or connector not coming undone when pulled).
- Switch the boat on and check IMS functionality : Start the IMS, check you hear the hydraulic pump running within 15 seconds of IMS startup, and, with boat free and clear to move, slightly operate the joysticks to check the track drive is operational.
- If charger was set to Force Charging, set back to Force Charging = False (in Main menu  / Settings / Master features / 4158 / Force Charging).
- Switch off boat.
- Close back the access hatch (cabin steps on Commuter, foredeck floor hatch on Original).

8.4 Emailing files to Iguana Yachts

- Retrieve both sets of .tr0 and .csv recorded files (IXXAT_canAnalyser3_MiniLog_date_time.tr0 and IXXAT_canAnalyser3_MiniLog_date_time.csv), namely :
  - the set recorded at end of charge + 5 hours.
  - And the set recorded at end of charge + 29 hours.

  and please kindly **email them over to Iguana after sales** for analysis with the battery manufacturer.

9 Post-processing files (for Iguana internal use)

- So as to convert them to .asc files. See dedicated instructions sheet.